A PROPER MAP OF SPACES QUASI-FINITE OVER ONE POINT IS FINITE ON A NEIGHBOURHOOD OF THAT POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7905-7916 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7918-7928, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin uncounted per ruling. Workhorse for L16232 in the same wave.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of Spaces, lemma locally-finite-type-quasi-finite-part; in-source lemma-characterize-finite.

Source reference. Original lemma 0ACK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7928–7928. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7848–7859. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7928–7928. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $y \in |Y|$. Assume*

(1) *f is proper, and*
(2) *f is quasi-finite at all $x \in |X|$ lying over y (Decent Spaces, Lemma 0ACK).*

Then there exists an open neighbourhood $V \subset Y$ of y such that $f|_{f^{-1}(V)} : f^{-1}(V) \to V$ is finite.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. By Morphisms of Spaces, Lemma 04NW the set of points at which f is quasi-finite is an open $U \subset X$. Let $Z = X \setminus U$. Then $y \notin f(Z)$. Since f is proper the set $f(Z) \subset Y$ is closed. Choose any open neighbourhood $V \subset Y$ of y with $Z \cap V = \emptyset$. Then $f^{-1}(V) \to V$ is locally quasi-finite and proper. Hence $f^{-1}(V) \to V$ is finite by Lemma 02LS. $\square$